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C. MANLEY
WOUND DRESSING
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Fig. 1.

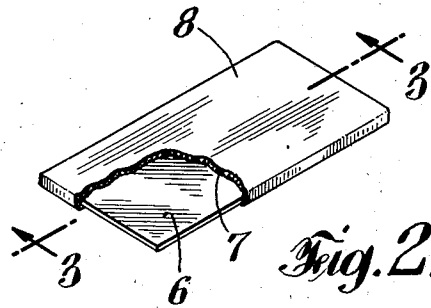
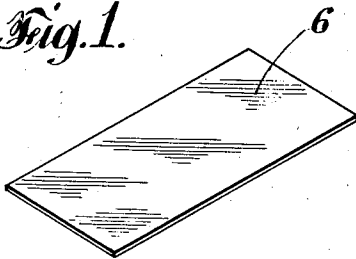


Fig. 2.



Fig. 3.

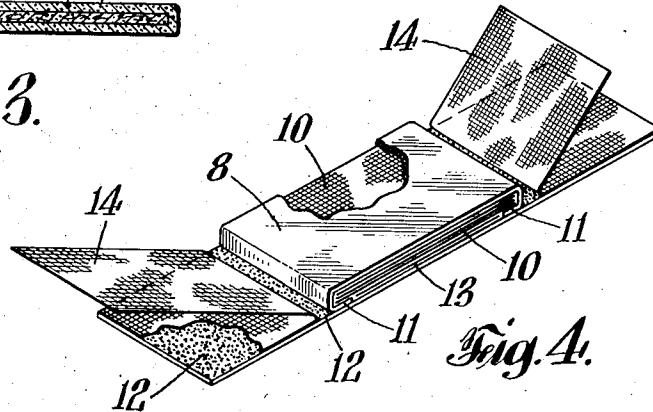


Fig. 4.

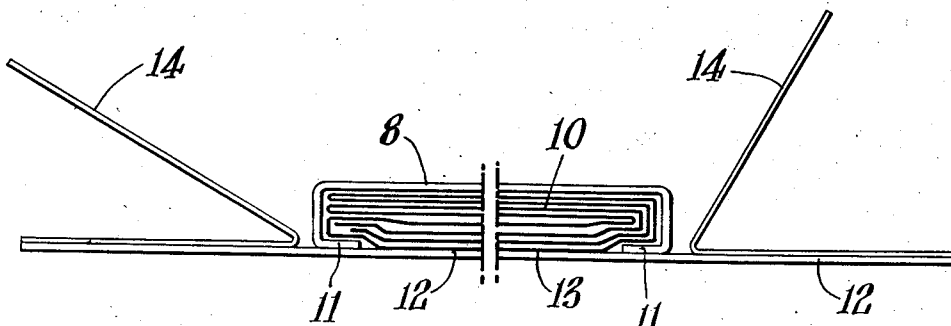


Fig. 5.

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WOUND DRESSING

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4 Claims. (Cl. 128—156)

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This invention relates to wound dressings and is particularly directed to dressings to be applied to wounds where a considerable amount of foreign substance such as coal dust has entered the wound before the dressing is applied.

Objects and advantages of the invention will be set forth in part hereinafter and in part will be obvious herefrom, or may be learned by practice with the invention, the same being realized and attained by means of the instrumentalities and combinations pointed out in the appended claims.

The invention consists in the novel parts, constructions, arrangements, combinations and improvements herein shown and described.

The accompanying drawings, referred to herein and constituting a part hereof, illustrate one embodiment of the invention, and together with the description, serve to explain the principles of the invention.

Of the drawings:

Fig. 1 is a view in perspective of a sheet of ordinary wax paper;

Fig. 2 is a view in perspective of the sheet of wax paper after it has been dipped several times in melted paraffin, which additional paraffin coating is shown partially cut away;

Fig. 3 is a cross section on the line 3—3 of the sheet shown in Fig. 2;

Fig. 4 is a perspective view of the complete wound dressing; and

Fig. 5 is a view in end elevation of the dressing shown in Fig. 4.

In coal mines dust is always present in the air and as a coating on all surfaces exposed to the air. When miners receive cuts, wounds or abrasions, loose coal dust enters the wound as it is present on the cutting or abrading surface as well as on the miners' skin, whenever these surfaces are exposed to the dusty atmosphere. Said dust, in addition to being a source of possible infection, is almost impossible to wash out of the wound and as a result the dark colored dust remains imbedded in the flesh after scar tissue has formed in the wound and it becomes healed. This results in a permanent discoloration of the area which is called a "blue mark" by those engaged in mining. Such "blue marks" or disfigurements are very common among miners and similar disfigurements occur in any occupation where similar dust conditions exist.

It is an object of my invention to provide a wound dressing which, when applied to a wound in which dust is imbedded, as described above, will facilitate healing of the wound and cleanse it of such dust particles so that permanent scar dis-

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coloration after the wound is healed is prevented. Another object of this invention is to provide such a wound dressing which may be easily and inexpensively manufactured. Another object is to provide a bandage which may be easily applied and which will offer maximum protection to the wound during the initial stage of healing when a protective clot is forming but which may be removed when such clot or scab has formed leaving behind, as part of the clot, a portion of its substance, which acts as the active cleansing agent as described above.

Still another object of this invention is to provide a bandage in which the active cleansing agent forms an integral part of the protective clot covering the wound. Said bandage so well protects the wound that it may be washed within a comparatively short time without said protective clot being dissolved, which would cause bleeding to recommence.

In general the invention provides a wound dressing comprising a strip of paraffin-impregnated paper or other readily impregnated film or sheet material on which the thickness of pure paraffin has been increased by repeated immersion of said strip into melted paraffin and subsequent withdrawal of said strip for cooling (or by other suitable coating operations) until the desired thickness of pure paraffin has been obtained; a series of layers of cloth gauze or other suitable absorbent material; and an adhesive tape backing upon which the gauze strips and the paraffin dipped paper are mounted so that the paraffin dipped paper makes primary contact with the wound. While paraffin has been found to be a useful and preferred substance, other waxes or like materials which are thermoplastic at low temperatures at or slightly below body or blood temperature, will be generally suitable for carrying out the major objects of the invention.

It will be understood that the foregoing general description and the following detailed description as well as exemplary and explanatory of the invention but are not restrictive thereof.

Referring to the drawings in detail and particularly to Figs. 1, 2 and 3, 6 is a conventional strip of wax or paraffin paper, 7 is a coat of paraffin which has been uniformly added to the surfaces of 6, as by repeated dipping into melted paraffin and subsequent withdrawal for cooling, 8 is the paraffin bandage element resulting from the deposit of paraffin layer 7 on paper 6.

Referring to Figs. 4 and 5 in detail, 10 is a strip of suitable absorbent material such as gauze or crinoline which in a preferred embodiment of the

invention is folded into a number of layers, 12 is a backing of adhesive tape and 14 is a strip of gauze or the like which is removed when the bandage is applied to the skin and which prevents adhesive surfaces from sticking to each other or other surfaces when the bandage is not in use.

Paraffin element 8, formed as above described, is preferably cut to the same width as cloth strip 10 which strip is folded upon itself as shown in Figs. 4 and 5. Overlapping ends of 8 are folded around folded strip 10 making contact at 11 with the adhesive surface of tape 12. The sticky surface of tape 12 contacts and adheres to strip 10 at 13 as shown in Figs. 4 and 5. Cloth strips 14 are adhered to overlapping ends of tape 12 as shown in Figs. 4 and 5 until such time as the bandage is to be used, when they are removed, thus exposing the adhesive surfaces of tape 12 which are then stuck to the skin on either side of the wound, holding the dressing firmly in place.

In operation paraffined element 8 directly contacts the wound and as a result the body heat melts or softens the paraffin which flows into the general conformation of the cut, forming a seal thereon. This paraffin seal actually becomes an integral part of the natural protective blood clot which forms and accomplishes the additional result of gathering up within itself loose particles of coal dust or the like. Within a short time, say two or three hours, depending on the size of the cut or wound a firm scab of blood fibrin and paraffin forms. Folded strip 10 has formed in the meantime a soft absorbent protective backing, the whole being firmly held in place by tape 12. After two or three hours, the bandage may be removed, with the paraffin-coated paper element 6 tearing away along the edges of the clot without reopening the wound to cause further bleeding. This leaves a firm, protective, air-tight covering on the cut which will enable the patient to wash the area without harm. Healing of the wound progresses at its normal rate and when the scab comes off at the end of this healing period, the dust and other particles will be imbedded in the paraffin-fibrin scab and not left in the scar tissue to permanently discolor it. It will be readily seen that suitable antiseptic and healing agents may be readily incorporated into the paraffin coating with consequent beneficial results. Although paraffin is used in the preferred embodiment, any suitable natural or synthetic wax like substance which is substantially molten at human body temperature may also be used.

This improved wound dressing which is of simple construction and which may be easily and cheaply manufactured will prove of great practical value in the prevention of permanent disfigurements which result when wounds are suffered by persons working in places where a dusty atmosphere prevails.

The invention in its broader aspects is not limited to the specific mechanisms shown and

described but departures may be made therefrom, within the scope of the accompanying claims, without departing from the principles of the invention and without sacrificing its chief advantages.

What I claim is:

1. A wound dressing comprising, in combination, a layer of paper, a plurality of layers of wax of uniform thickness attached to the surfaces of said paper, said wax having a melting point below body or blood temperature, a plurality of layers of gauze, said waxed paper being folded around said layers of gauze so as to enclose said layers, and a strip of adhesive backing attached to the folded ends of said waxed paper and providing a support for said paper and gauze.

2. A wound dressing comprising, in combination, a layer of paper, a plurality of layers of thermoplastic sealing material of uniform thickness attached to the surfaces of said paper, said thermoplastic material having a melting point below body or blood temperature, a plurality of layers of gauze, said paper being folded around said layers of gauze so as to enclose said layers, and a strip of adhesive backing attached to the folded ends of said paper and providing a support for said paper and gauze.

3. A wound dressing, including in combination a bandage of absorbent material, and a layer of thermoplastic sealing material of uniform thickness covering said absorbent material along its entire outer surface, said thermoplastic layer adapted to be in direct contact with the wound and being continuous to form a protective airtight covering and having a melting point below body or blood temperature whereby said thermoplastic layers melts on body contact to seal the wound and pick up foreign particles therein, and an adhesive layer external to the bandage for attachment to the skin of the user.

4. A dressing as specified in claim 2, wherein said thermoplastic layer comprises a sheet of paper with a uniform thickness of paraffin thereon, said paraffin having a melting point below blood or body temperatures.

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